Vijay Academy Presents

Savitribai Phule Pune University (SPPU)

Data Science and Big Data Analytics (DSBDA)

TE Computer Engineering (2019 Pattern)

One Shot Lecture

Unit 5. Big Data Analytics and Model Evaluation

What You Get : Strategy For Exam :

- Simple Explanation
- PPT + Notes
- PYQs + Most IMP Questions

1: K-Means Clustering

What is K-Means Clustering?

K-Means Clustering is an **unsupervised machine learning algorithm** that divides data into **K number of clusters** based on similarity.

Simply put — *grouping similar data points together into K groups automatically.*

Key Terms:

Term	Meaning
K	Number of clusters to form
Centroid	Center point of each cluster
Cluster	Group of similar data points
Euclidean Distance	Distance between two points

Euclidean Distance Formula:

Distance = $\sqrt{(x2-x1)^2 + (y2-y1)^2}$

1: K-Means Clustering

Choose K

↓

Initialize K Centroids randomly

↓

Calculate distance of each point
from all centroids

↓

Assign each point to nearest centroid

↓

Recalculate centroid of each cluster

↓

Did centroids change?

↓YES ↓NO

Repeat STOP (Converged)

Complete Numerical Example (From PYQ 2024):

Q5) a) Suppose you have the following dataset containing the coordinates of points in a 2-dimensional space: **[9]**

Point	X Coordinate	Y Coordinate
A	2	3
B	4	7
C	3	5
D	6	9
E	8	6
F	7	8

Perform K-means clustering on this dataset with K = 2. Assume the initial centroids to be (2,3) and (8,6). Compute the new centroids after each iteration until convergence, and assign points to their nearest centroids.

Ans :

Problem: Perform K-Means with K=2

Points: A(2,3), B(4,7), C(3,5), D(6,9), E(8,6), F(7,8)

Initial Centroids: C1=(2,3), C2=(8,6)

Iteration 1:

Calculate distances from C1(2,3) and C2(8,6):

Point	Dist from C1(2,3)	Dist from C2(8,6)	Cluster
A(2,3)	$\sqrt{(2-2)^2+(3-3)^2}$ = **0**	$\sqrt{(8-2)^2+(6-3)^2}$ = 6.7	C1
B(4,7)	$\sqrt{(4-2)^2+(7-3)^2}$ = 4.47	$\sqrt{(8-4)^2+(6-7)^2}$ = **4.12**	C2
C(3,5)	$\sqrt{(3-2)^2+(5-3)^2}$ = **2.24**	$\sqrt{(8-3)^2+(6-5)^2}$ = 5.1	C1
D(6,9)	$\sqrt{(6-2)^2+(9-3)^2}$ = 7.21	$\sqrt{(8-6)^2+(6-9)^2}$ = **3.6**	C2
E(8,6)	$\sqrt{(8-2)^2+(6-3)^2}$ = 6.7	$\sqrt{(8-8)^2+(6-6)^2}$ = **0**	C2
F(7,8)	$\sqrt{(7-2)^2+(8-3)^2}$ = 7.07	$\sqrt{(8-7)^2+(6-8)^2}$ = **2.24**	C2

Cluster 1: A(2,3), C(3,5)   **Cluster 2:** B(4,7), D(6,9), E(8,6), F(7,8)

Recalculate Centroids:

New C1 = Mean of A(2,3) and C(3,5)

= ((2+3)/2, (3+5)/2)

= (2.5, 4)

New C2 = Mean of B(4,7), D(6,9), E(8,6), F(7,8)

= ((4+6+8+7)/4, (7+9+6+8)/4)

= (25/4, 30/4)

= (6.25, 7.5)

Iteration 2:

Calculate distances from C1(2.5,4) and C2(6.25,7.5):

Point	Dist from C1(2.5,4)	Dist from C2(6.25,7.5)	Cluster
A(2,3)	$\sqrt{(0.25+1)} =$ **1.12**	$\sqrt{(17.56+20.25)} = 6.15$	C1
B(4,7)	$\sqrt{(2.25+9)} =$ **3.35**	$\sqrt{(5.06+0.25)} = 2.3$	C2
C(3,5)	$\sqrt{(0.25+1)} =$ **1.12**	$\sqrt{(10.56+6.25)} = 4.1$	C1
D(6,9)	$\sqrt{(12.25+25)} =$ **6.1**	$\sqrt{(0.06+2.25)} = 1.52$	C2
E(8,6)	$\sqrt{(30.25+4)} =$ **5.85**	$\sqrt{(3.06+2.25)} = 2.3$	C2
F(7,8)	$\sqrt{(20.25+16)} =$ **6.02**	$\sqrt{(0.56+0.25)} = 0.9$	C2

Cluster 1: A(2,3), C(3,5) → Same as before!

Cluster 2: B(4,7), D(6,9), E(8,6), F(7,8) → Same as before!

Centroids did not change

Final Result:

Cluster 1: A(2,3), C(3,5) → Centroid (2.5, 4)

Cluster 2: B(4,7), D(6,9), E(8,6), F(7,8) → Centroid (6.25, 7.5)

Thank You!

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